Heterogeneity and Incentives in Dynamic Matching Markets: Outside Options, Liquidity, and Competition between Exchanges

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June 9, 2026

Abstract

We extend the two-type continuous-time dynamic matching model to allow easy-to-match (E) agents to face outside-option pressure: E agents become critical (depart) at rate $\rho \geq 1$ rather than rate 1. This models altruistic donors or versatile workers who can exit the market when attractive alternatives appear elsewhere. We derive closed-form loss functions for all four canonical policies (Greedy, Patient, Type-Aware Greedy, Type-Aware Patient) as functions of ρ and d_1 (H-E compatibility density), revealing a fundamental asymmetry: **Patient-type policies are directly exposed to outside-option pressure while Greedy-type policies are not, at leading order.**

Key results: $L^{\text{Patient}}(d_1, \rho) = 4\rho W(d_1/(8\rho))/d_1$ and $L^{\text{TA-Greedy}}(d_1, \rho) = 2W(d_1/4)/d_1$ (independent of ρ), yielding the duality $L^{\text{Patient}}(d_1, \rho) = L^{\text{TA-Greedy}}(d_1/(2\rho))$. For TA-Patient the exponential decay rate becomes $e^{-d_1/(4(1+\rho))}$: stronger outside options compress the exponential advantage. We analyze competition between platforms, showing private exchanges can dominate a public monopoly when markets are thick enough. These findings speak to the empirical debate over kidney-exchange consolidation.

Contents

1 Introduction

A central question in matching-market design is whether to centralize or fragment. In kidney exchange, the US relies on multiple hospital-run consortia—the National Kidney Registry (NKR, daily clearings), the Alliance for Paired Kidney Donation (daily or on new-entry), and the OPTN clearinghouse (weekly)—alongside hospital-specific programs that match locally before accessing national pools. The economic literature has emphasized gains from thicker markets: pooling more agents raises match rates, especially for hard-to-match types. Yet fragmentation persists and in some cases improves outcomes.

This paper identifies a mechanism for why: when easy-to-match agents—altruistic donors, bridge donors, versatile workers—face outside options that cause them to leave the market, the key design variable is not pool size alone but how quickly the market captures value from E agents before they exit. A platform that credibly reduces easy agents' effective departure rate (through priority credits, better match quality, or commitment mechanisms) delivers strictly higher welfare for hard-to-match agents, and private platforms unconstrained by public procurement rules may have a comparative advantage on this dimension.

The heterogeneous model. We model the market as a continuous-time stochastic process with two agent types: hard-to-match (H) and easy-to-match (E). The compatibility graph on the pool is a Stochastic Block Model: H-H edges are forbidden, H-E edges appear independently with probability $d_1/(2m)$, and E-E edges with probability $d_2/(2m)$, where $d_2 > d_1 > 0$ captures heterogeneity. This is the *sparse* regime: compatibility probabilities vanish as $1/m$, so the expected H-E degree of a pooled agent is $O(1)$, unlike the dense regime of Ashlagi, Nikzad, and Strack (2023) where p_{HE} is a fixed constant. Empirically, Liu et al. (2018) find in DiDi carpooling data that moderate heterogeneity favors Patient, but pronounced heterogeneity can reverse this advantage—consistent with the crossover behavior our model predicts.

The key parameter: ρ. We introduce the E departure rate $\rho \geq 1$. When $\rho = 1$, E agents depart no faster than H agents (baseline). When $\rho > 1$, E agents become critical faster. The steady-state E pool size is proportional to $1/\rho$, so the inventory of potential rescue partners for H agents shrinks as ρ rises.

Main findings.

1. *Greedy policies are ρ-insensitive at leading order.* Neither Greedy nor TA-Greedy loss depends on ρ as $d_2 \to \infty$. The H-E density d_1 is the binding constraint.
2. *Patient loss has an exact closed form in ρ:* $L^{\text{Patient}}(d_1, \rho) = 4\rho W(d_1/(8\rho))/d_1$, where W is the Lambert W function.
3. *TA-Patient's exponential advantage erodes with ρ.* The decay exponent decreases from $d_1/8$ (at $\rho = 1$) to $d_1/(4(1 + \rho))$.
4. *Duality.* $L^{\text{Patient}}(d_1, \rho) = L^{\text{TA-Greedy}}(d_1/(2\rho))$: outside-option pressure on E is equivalent to a rescaling of H-E density.
5. *Competition can be welfare-improving.* A private platform that uses TA-Patient and reduces ρ dominates a public Patient platform in thick markets.

Remark 1 (Notation conflict)**.** In prior work on asymmetric markets, ρ sometimes denotes the arrival-rate ratio between types. Here ρ is exclusively the E departure rate; arrival rates remain equal at $m/2$ each.

2 Model

2.1 Setup

The market has two types: hard-to-match (H) and easy-to-match (E). Both arrive as Poisson processes at rate $m/2$ each (total rate m).

Compatibility. Any two agents in the pool are independently compatible with type-pair probabilities:

$$p_{HH} = 0, \quad p_{HE} = p_{EH} = \frac{d_1}{2m}, \quad p_{EE} = \frac{d_2}{2m}, \qquad d_2 > d_1 > 0.$$

Departure rates. H agents become critical at rate 1; E agents become critical at rate $\rho \geq 1$. A critical agent must be matched immediately or it perishes.

Pool state. Let $N_t = (H_t, E_t)$ denote the pool state. In rescaled variables $x = H_t/m$ and $y = E_t/m$, the pool evolves via arrivals, criticalities, and matches.

Neighbor probabilities. Given pool state (h, e), the probability that a given H agent has at least one compatible (E) neighbor is:

$$\pi_H(h, e) = 1 - \left(1 - \tfrac{d_1}{2m}\right)^e \approx 1 - e^{-d_1 y/2},$$

and the probability that a given E agent has at least one compatible (H or E) neighbor is:

$$\pi_E(h, e) = 1 - \left(1 - \tfrac{d_1}{2m}\right)^h \left(1 - \tfrac{d_2}{2m}\right)^e \approx 1 - e^{-(d_1 x + d_2 y)/2}.$$

2.2 Four policies

We study four canonical policies:

- **Greedy**: match on arrival, choosing uniformly among compatible neighbors in the pool.
- **TA-Greedy**: match on arrival, prioritizing H-type neighbors for E arrivals.
- **Patient**: match at criticality, choosing uniformly among compatible neighbors in the pool.
- **TA-Patient**: match at criticality, prioritizing H-type neighbors for E criticals.

Under any Greedy variant the pool is edgeless: arriving agents have not yet encountered each other, so no two pooled agents have a tested edge. Under any Patient variant the pool retains a random graph on pooled agents.

A fifth policy, **TA-Hybrid** (match H immediately upon receiving a compatible E neighbor; be patient otherwise), interpolates between TA-Greedy and TA-Patient and is left for future analysis.

2.3 Drift and mean-field fixed point

For a given policy, the drift of (H_t, E_t) is:

$$f_H(h, e) = \mathbb{E}[\Delta H_t \mid H_t = h, E_t = e], \qquad f_E(h, e) = \mathbb{E}[\Delta E_t \mid H_t = h, E_t = e].$$

A *mean-field fixed point* (x^*, y^*) satisfies $f_H(mx^*, my^*) = 0$ and $f_E(mx^*, my^*) = 0$. Under standard concentration arguments (supported numerically by occupancy heatmaps that show the stationary distribution concentrating tightly near (h^*, e^*)), the long-run loss is well approximated by evaluating at the fixed point.

2.4 Loss

The long-run loss is the fraction of agents who perish unmatched:

$$L = \lim_{T \to \infty} \frac{\mathbb{E}[\text{agents perishing unmatched by } T]}{\mathbb{E}[\text{total arrivals by } T]}. \tag{1}$$

In mean-field terms, at the fixed point (x^*, y^*):

$$L = x^*(1 - \pi_H^*) + \rho\, y^*(1 - \pi_E^*), \tag{2}$$

where $\pi_H^* = 1 - e^{-d_1 y^*/2}$ and $\pi_E^* = 1 - e^{-(d_1 x^* + d_2 y^*)/2}$. The two terms are the H-perish rate (H becomes critical without a compatible neighbor) and the E-perish rate (E becomes critical at rate ρ without any compatible neighbor).

3 Analysis

3.1 Greedy: leading-order independence from ρ

Under Greedy, the drift of the rescaled pool (x, y) is:

$$F_H(x,y) = \tfrac{1}{2}(1-\pi_H) - x - \tfrac{1}{2}\,r_{EH}, \tag{3}$$

$$F_E(x,y) = \tfrac{1}{2}(1-\pi_E) - \rho y - \tfrac{1}{2}\,\pi_H - \tfrac{1}{2}\,r_{EE}, \tag{4}$$

where r_{EH} and r_{EE} are the probabilities that an arriving E agent matches an H or E agent, respectively.

Lemma 1 (Greedy loss is ρ-independent at leading order)**.** ***[Proved]*** *As $d_2 \to \infty$, the Greedy fixed-point x^* is the unique root in $(0, 1/2)$ of*

$$g_{d_1}(x) := x \exp\left(\frac{d_1 x(1-x)}{2(1-2x)}\right) = 1,$$

independent of ρ. Hence $L^{\text{Greedy}}(d_1, \rho) = x^*(d_1) + O(1/d_2)$.

Proof. In rescaled-degree variables with $\pi_H = 1 - e^{-d_1 y/2}$ and similarly for π_E and r_{EE}, the fixed-point equations become:

$$x + \rho y = e^{-d_1 y/2} + e^{-(d_1 x + d_2 y)/2} - 1, \tag{S1}$$

$$x - \rho y = \left(1 - e^{-(d_1 x + d_2 y)/2}\right) \cdot \frac{d_2 y}{d_1 x + d_2 y}. \tag{S2}$$

Step 1: Scaling of y. Write $\rho y = c/d_2$ for $c = c(x, d_1) > 0$. Substituting into (S2) and dropping the $O(1/d_2)$ term on the left side:

$$x = \left(1 - e^{-(d_1 x + c)/2}\right) \cdot \frac{c}{d_1 x + c}.$$

Setting $\alpha = d_1 x + c$ and rearranging: $x\alpha = c(1 - e^{-\alpha/2})$, which expands to $d_1 x^2 + xc = c - ce^{-\alpha/2}$. At leading order (the exponential term is $O(e^{-c/2})$ and $c \to \infty$ as $d_2 \to \infty$):

$$c = \frac{d_1 x^2}{1 - 2x}. \tag{$*$}$$

This relation is independent of ρ.

Step 2: Scalar equation for x. Substituting $y = c/(d_2 \rho)$ into (S1) and taking $d_2 \to \infty$ (so $e^{-d_1 y/2} \to 1$ and $e^{-(d_1 x + c)/2}$ is the surviving exponential):

$$x \;\longrightarrow\; e^{-(d_1 x + c)/2}.$$

Substituting ($*$):

$$x = \exp\left(-\frac{d_1 x}{2} - \frac{d_1 x^2}{2(1-2x)}\right) = \exp\left(-\frac{d_1 x(1-x)}{2(1-2x)}\right),$$

which is $g_{d_1}(x) = 1$, independent of ρ.

Step 3: Loss. Since $\rho y = c/d_2 \to 0$ and $1 - \pi_E = e^{-(d_1 x + d_2 y)/2} \to 0$, we have $\rho y(1 - \pi_E) = O(1/d_2)$. Thus $L = x(1 - \pi_H) + O(1/d_2) = x^*(d_1) + O(1/d_2)$, independent of ρ. □

Remark 2 (Intuition)**.** Under Greedy, matching happens at agent arrival. When d_2 is large, the E pool is dense enough to serve as a reliable standing inventory for H arrivals regardless of how quickly E agents depart. The binding constraint is the H-E compatibility density d_1; the E departure rate is a second-order correction.

3.2 TA-Greedy: same leading-order independence

Under TA-Greedy, an arriving E agent with both H and E compatible neighbors prioritizes the H neighbor. At leading order as $d_2 \to \infty$, the fixed-point analysis runs as in Lemma 1 with one modification: the H-E compatibility density is effectively doubled (each E arrival rescues the H neighbor immediately), which shifts the equation for x^*. The outcome is:

$$L^{\text{TA-Greedy}}(d_1, \rho) = \frac{2W(d_1/4)}{d_1} + O(1/d_2),$$

independent of ρ at leading order.

Sketch. The key step is identical to Step 1 in the Greedy proof: the E pool satisfies $\rho y = O(1/d_2)$, so ρ enters the fixed-point equations only through the vanishing term $\rho y = c/d_2$. The scalar equation for x^* that survives the $d_2 \to \infty$ limit is ρ-free. The Lambert-W formula $2W(d_1/4)/d_1$ comes from applying the same argument with H-priority, which changes the exponent in g_{d_1} by a factor of 2.

3.3 Patient: closed-form ρ-dependence

Under Patient, agents wait in the pool until critical, then match uniformly among compatible neighbors.

Lemma 2 (Patient loss with ρ)**.** ***[Proved]*** *As $d_2 \to \infty$, the Patient fixed point satisfies $x^* \to 1/2$ and $y^* = y^*(d_1, \rho)$ where $e^{-d_1 y^*/2} = 4\rho y^*$. The loss is:*

$$\boxed{L^{\text{Patient}}(d_1, \rho) = \frac{4\rho\, W(d_1/(8\rho))}{d_1}.}$$

Proof. At the mean-field fixed point, the steady-state conditions for the H and E pool sizes are:

$$\tfrac{1}{2} = x + \rho y\, \pi_E\, \rho_{E\to H}, \tag{P1}$$
$$\tfrac{1}{2} = x\, \pi_H + \rho y \big(1 + \pi_E\, \rho_{E\to E}\big). \tag{P2}$$

In (P1): H flows in at rate $1/2$; it leaves at rate x (all critical H, whether matched or not) plus rate $\rho y\, \pi_E \rho_{E\to H}$ (H matched by a critical E). In (P2): E flows in at rate $1/2$; it leaves at rate $x\pi_H$ (E matched by critical H) plus rate $\rho y(1 + \pi_E \rho_{E\to E})$ (each critical E removes itself, and with probability $\pi_E \rho_{E\to E}$ it also removes a second E as the matched partner).

Step 1: $x \to 1/2$ as $d_2 \to \infty$. As $d_2 \to \infty$ with a positive E pool, critical E agents have almost surely at least one compatible neighbor, but the fraction of those neighbors that are H-type vanishes: $d_1 x/(d_1 x + d_2 y) \to 0$. Under type-unaware Patient, E matches uniformly, so $\pi_E \rho_{E\to H} \to 0$. From (P1): $1/2 = x + o(1)$, hence $x \to 1/2$.

Step 2: Scalar equation for y. With $d_2 \to \infty$, every critical E almost surely finds an E-compatible neighbor, so $\pi_E \rho_{E\to E} \to 1$ and thus $1 + \pi_E \rho_{E\to E} \to 2$. Also $\pi_H = 1 - e^{-d_1 y/2}$. Substituting $x = 1/2$ in (P2):

$$\tfrac{1}{2} = \tfrac{1}{2}\big(1 - e^{-d_1 y/2}\big) + 2\rho y.$$

Simplifying:

$$e^{-d_1 y/2} = 4\rho y. \tag{5}$$

Step 3: Lambert W form. Multiply both sides of (5) by $e^{d_1 y/2}$: $1 = 4\rho y\, e^{d_1 y/2}$. Set $u = d_1 y/2$, so $y = 2u/d_1$:

$$1 = \frac{8\rho}{d_1}\, u\, e^u \implies u\, e^u = \frac{d_1}{8\rho}.$$

By definition of the Lambert W function, $u = W(d_1/(8\rho))$, giving $y^* = 2W(d_1/(8\rho))/d_1$.

Step 4: Loss formula. Adding (P1) and (P2) and using $\rho_{E\to H} + \rho_{E\to E} = 1$ (a critical E with at least one compatible neighbor matches exactly one agent), so that $\pi_E \rho_{E\to H} + \pi_E \rho_{E\to E} = \pi_E$:

$$1 = x(1 + \pi_H) + \rho y(1 + \pi_E).$$

From $L = x(1-\pi_H) + \rho y(1-\pi_E)$ and the equation above:

$$L = (x+\rho y) - (x\pi_H + \rho y \pi_E) = (x+\rho y) - \big(1-(x+\rho y)\big) = 2(x+\rho y) - 1.$$

With $x = 1/2$ and $y^* = 2W(d_1/(8\rho))/d_1$:

$$L = 2\left(\tfrac{1}{2} + \rho \cdot \frac{2W(d_1/(8\rho))}{d_1}\right) - 1 = \frac{4\rho\, W(d_1/(8\rho))}{d_1}. \qquad \square$$

Remark 3 (Behavior in ρ)**.** As $\rho \to 1$: recover the baseline Patient loss. As $\rho \to \infty$: the argument of W tends to 0, so $W(d_1/(8\rho)) \approx d_1/(8\rho)$, giving $L \to 1/2$ (half of all agents perish). As $\rho \to 0$: E agents are permanently sticky and $L \to 0$. The log-elasticity is $\partial \log L/\partial \log \rho = W(d_1/(8\rho))/(1 + W(d_1/(8\rho))) \in (0,1)$.

3.4 TA-Patient: identities and exponential decay

Under TA-Patient, critical E agents match their H-compatible neighbor first; only if no H neighbor exists do they match an E.

Lemma 3 (TA-Patient fixed-point identities)**.** *[Proved] Any solution (x,y) of the TA-Patient fixed-point system (as $d_2 \to \infty$) satisfies:*

$$L^{\text{TA-Patient}} = x\, e^{-d_1 y/2}, \tag{6}$$

$$x\, e^{-d_1 y/2} = 2\rho y\, e^{-d_1 x/2}. \tag{7}$$

Proof. As $d_2 \to \infty$, the TA-Patient fixed-point equations are:

$$\tfrac{1}{2} = x + \rho y\big(1 - e^{-d_1 x/2}\big), \tag{TAP1}$$

$$\tfrac{1}{2} = x\big(1 - e^{-d_1 y/2}\big) + \rho y\big(1 + e^{-d_1 x/2}\big). \tag{TAP2}$$

In (TAP1): all critical H leave at rate x; critical E's match H with probability $1 - e^{-d_1 x/2}$ (probability of having at least one H neighbor) at rate ρy. In (TAP2): critical H's match E with probability $\pi_H = 1 - e^{-d_1 y/2}$ at rate x; critical E's always leave at rate ρy (the $+1$ term), and additionally pull a second E from the pool when they find no H neighbor and instead match an E (the $+e^{-d_1 x/2}$ term, since $d_2 \to \infty$ guarantees an E neighbor whenever needed).

Step 1: Summing (TAP1) and (TAP2).

$$\begin{aligned} 1 &= \big[x + \rho y(1 - e^{-d_1 x/2})\big] + \big[x(1 - e^{-d_1 y/2}) + \rho y(1 + e^{-d_1 x/2})\big] \\ &= 2x + 2\rho y - x e^{-d_1 y/2} \underbrace{- \rho y e^{-d_1 x/2} + \rho y e^{-d_1 x/2}}_{=0}. \end{aligned}$$

Hence $L = 2(x+\rho y) - 1 = x e^{-d_1 y/2}$ (using the same aggregate-balance identity as in the Patient proof, which holds because TA-Patient also satisfies $\pi_E \rho_{E\to H} + \pi_E \rho_{E \to E} = \pi_E$).

Step 2: Subtracting (TAP1) from (TAP2).

$$\begin{aligned} 0 &= \big[x(1 - e^{-d_1 y/2}) + \rho y(1 + e^{-d_1 x/2})\big] - \big[x + \rho y(1 - e^{-d_1 x/2})\big] \\ &= -x e^{-d_1 y/2} + 2\rho y e^{-d_1 x/2}. \end{aligned}$$

Thus $x e^{-d_1 y/2} = 2\rho y e^{-d_1 x/2}$. $\square$

Conjecture 1 (TA-Patient exponential decay with ρ)**.** *For $d_1 \to \infty$ at fixed $\rho \geq 1$:*

$$\boxed{L^{\text{TA-Patient}}(d_1, \rho) \;\sim\; \frac{(2\rho)^{1/(1+\rho)}}{2(1+\rho)}\, e^{-d_1/(4(1+\rho))}.}$$

The conjecture is supported by numerical simulation. A heuristic derivation: equations (6)–(7) together with (TAP1) suggest the saddle-point scaling $y/x \to 1/\rho$ for large d_1, leading to the linear system $d_1/(4(1+\rho))$ in the exponent and the prefactor $(2\rho)^{1/(1+\rho)}/(2(1+\rho))$ from matching boundary conditions.

The decay exponent $A(\rho) = 1/(4(1+\rho))$ is strictly decreasing in ρ:

ρ	0.5	1	2	5	10	20
$A(\rho)$	1/6	1/8	1/12	1/24	1/44	1/84

Doubling ρ from 1 to 2 requires 50% more H-E compatibility density to achieve the same loss level.

4 Duality and Policy Comparisons

Lemma 4 (ρ-generalized Patient–TA-Greedy duality)**.** ***[Proved]***

$$L^{\text{Patient}}(d_1, \rho) \;=\; L^{\text{TA-Greedy}}\left(\frac{d_1}{2\rho}\right).$$

Proof. From Lemma 2: $L^{\text{Patient}}(d_1, \rho) = 4\rho W(d_1/(8\rho))/d_1$. From Section 3.2: for any $d_1' > 0$, $L^{\text{TA-Greedy}}(d_1') = 2W(d_1'/4)/d_1'$. Setting $d_1' = d_1/(2\rho)$:

$$L^{\text{TA-Greedy}}\left(\frac{d_1}{2\rho}\right) = \frac{2W\big(d_1/(8\rho)\big)}{d_1/(2\rho)} = \frac{4\rho\, W(d_1/(8\rho))}{d_1} = L^{\text{Patient}}(d_1, \rho). \qquad \square$$

Remark 4 (Economic interpretation)**.** Outside-option pressure on E (higher ρ) is equivalent, for Patient policy, to reducing the H-E compatibility density by a factor of 2ρ. A Patient planner facing E departure rate ρ achieves the same loss as a TA-Greedy planner in a market where d_1 has been divided by 2ρ. Conversely, halving ρ delivers the same loss improvement as doubling d_1.

Policy ranking. For large d_1 (thick, heterogeneous markets), the loss ordering is:

$$L^{\text{TA-Patient}} \;\ll\; L^{\text{TA-Greedy}} \;\approx\; L^{\text{Greedy}} \;<\; L^{\text{Patient}},$$

with TA-Patient achieving exponential loss and the others polynomial. Importantly, *without type-aware selection, being Patient is worse than being Greedy*: under type-unaware Patient, the abundant E pool is consumed by E-E matches at criticality rather than by H rescues. Patience helps only when the matching rule deliberately directs critical E agents toward H.

For thin markets (d_1 small), the ranking can differ. In particular, type-unaware Patient can be dominated by Greedy because waiting does not build enough matching opportunities for H—the H pool is too sparse for Patient's E-rescue mechanism to operate.

Threshold ρ^*. Numerical simulations (Figure: policy performance vs. E departure rate) reveal a threshold $\rho^*(d_1, d_2)$ above which the Patient and TA-Patient advantage over Greedy-type policies vanishes. For $\rho > \rho^*$, E turnover is fast enough that the E inventory is too thin to benefit from waiting. Characterizing ρ^* analytically is an open problem.

5 Competition Between Exchanges

5.1 Two-platform model

Consider two platforms serving the same population of H agents but competing for E agents:

- **Platform A** (public): uses Patient policy; attracts fraction $\alpha \in (0, 1)$ of E agents; E departure rate ρ_0.
- **Platform B** (private): uses TA-Patient policy; attracts fraction $1 - \alpha$ of E agents; E departure rate $\rho_B < \rho_0$ (reduced by platform-specific retention mechanisms).

Platform B's ability to offer $\rho_B < \rho_0$ reflects tools unavailable to the public platform: priority credits for altruistic donors, selective offering of younger or better-matched recipients, and commitment mechanisms that bind donors to the pool.

The composite E departure rate seen by H agents across both pools is approximated by the harmonic mean:

$$\rho_{\text{eff}}(\alpha) = \left[\frac{\alpha}{\rho_0} + \frac{1-\alpha}{\rho_B}\right]^{-1}. \tag{8}$$

As α decreases (more E agents migrate to the private platform), ρ_{eff} falls: the system-wide E pool turns over more slowly.

5.2 Welfare comparison

Proposition 1 (Competition can be welfare-improving)**.** ***[Open / numerical]*** *For large d_1:*

$$L^{\text{TA-Patient}}\big(d_1,\, \rho_{\text{eff}}(\alpha)\big) \;<\; L^{\text{Patient}}(d_1, \rho_0),$$

because exponential decay (TA-Patient) dominates polynomial decay (Patient) as $d_1 \to \infty$, even after the fragmentation penalty from $\alpha < 1$. For moderate d_1, welfare depends on a threshold $d_1^*(\rho_0, \rho_B, \alpha)$.

Remark 5 (Empirical context)**.** In the US kidney exchange system, the NKR (daily), the Alliance for Paired Kidney Donation (daily or on new-entry), and the OPTN clearinghouse (weekly) operate at different clearing frequencies, which correspond to different effective E departure rates. Hospital programs like UC's that match locally before accessing national pools effectively run TA-Patient-like protocols. Our model predicts that this layered, fragmented structure need not be welfare-reducing: if faster-clearing platforms also implement type-aware protocols and invest in donor retention (lower ρ), the TA-Patient exponential advantage can outweigh the cost of pool fragmentation in thick enough markets.

6 Discussion

Arrival imbalance. When E arrivals exceed H arrivals (fraction $\lambda_E > 1/2$ of total arrivals), the TA-Patient advantage is amplified: the deeper E pool strengthens the rescue channel for H agents. Numerical results show that sufficient imbalance (λ_E close to 1) can preserve the TA-Patient advantage even at high ρ, because a thicker E pool compensates for rapid E turnover. This suggests a complementary design lever: platforms can sustain the TA-Patient advantage under high ρ by actively recruiting disproportionately more E-type participants.

Finite-d_2 corrections. The $O(1/d_2)$ corrections to Greedy and TA-Greedy do depend on ρ: when d_2 is finite, E-E matching becomes imperfect and E-pool dynamics interact with ρ. The crossover density d_2^* at which policy rankings flip in ρ is an open numerical question.

Formal competition model. The two-platform welfare comparison relies on the effective-rate approximation (8), which treats the composite system as a single pool with blended E departure rate. A rigorous treatment would model the joint Markov chain on $(H_t^A, E_t^A, H_t^B, E_t^B)$ and derive the coupled mean-field fixed-point equations.

Stationary concentration. The mean-field approach requires the stationary distribution of the CTMC to concentrate near the fixed point (x^*, y^*). This is supported numerically by occupancy heatmaps that show tight mass near (h^*, e^*), and is an extension of the concentration arguments in the baseline AKL paper.

7 Conclusion

The E departure rate ρ introduces a fundamental asymmetry across policies: Patient-type policies are ρ-sensitive while Greedy-type policies are not at leading order. The closed form $L^{\text{Patient}}(d_1, \rho) = 4\rho W(d_1/(8\rho))/d_1$ makes this precise, and the duality $L^{\text{Patient}}(d_1, \rho) = L^{\text{TA-Greedy}}(d_1/(2\rho))$ translates it into a concrete equivalence: a unit increase in $\log \rho$ is as damaging to Patient performance as a factor-2ρ decrease in H-E compatibility density.

For TA-Patient—the exponentially superior policy—higher ρ compresses the decay exponent from $d_1/8$ (at $\rho = 1$) to $d_1/(4(1+\rho))$. A platform that retains E agents gains exponentially in thick markets. The competition model shows that fragmentation is not uniformly harmful: a private platform that sufficiently reduces ρ and uses TA-Patient matching can dominate a public Patient platform once markets are thick enough.

Retention incentives. The analysis points to three concrete mechanisms for reducing ρ:

1. *Priority credits*: E agents who donate accumulate credits for priority matching in the future, making staying in the exchange more attractive than outside options.
2. *Quality matching*: Private exchanges can offer younger or healthier recipients to altruistic donors—improving the inside option relative to the outside option and directly reducing ρ.
3. *Commitment mechanisms*: Soft commitments (listing the donor as active) or hard commitments (binding agreements with the exchange) reduce turnover directly.

Each of these shifts the effective ρ downward, and our results quantify exactly how much: reducing ρ by a factor of 2 is equivalent to doubling d_1, i.e., halving the fraction of incompatible pairs in the market.